EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

EM Wave

1. Microwave oven acts on the principle of transferring electrons from lower to
(A) higher energy levels in water molecule
(B) giving rotational energy to water molecules
(C) giving translational energy to water molecules
(D) giving vibrational energy to water molecules

PWO081

2. A plane electromagnetic wave having a frequency $\nu = 23.9\ GHz$ propagates along the positive z-direction in free space. The peak value of the electric field is 60 V/m. Which among the following is the acceptable magnetic field component in the electromagnetic wave?
(A) $\vec{B} = 2\times10^{7}\sin(0.5\times10^{3}z + 1.5\times10^{11}t)\hat{i}$
(B) $\vec{B} = 2\times10^{-7}\sin(1.5\times10^{2}x + 0.5\times10^{11}t)\hat{j}$
(C) $\vec{B} = 2\times10^{-7}\sin(0.5\times10^{3}z - 1.5\times10^{11}t)\hat{i}$
(D) $\vec{B} = 60\sin(0.5\times10^{3}x + 1.5\times10^{11}t)\hat{k}$

PWO082

3. The electric field of a plane polarized electromagnetic wave in free space at time $t = 0$ is given by an expression $\vec{E}(x,y) = 10\hat{j}\cos[(6x + 8z)]$. The magnetic field $\vec{B}(x, z, t)$ is given by:
(c is the velocity of light)

(A) $\frac{1}{c}(6\hat{k} + 8\hat{i})\cos[(6x - 8z + 10ct)]$ (B) $\frac{1}{c}(6\hat{k} - 8\hat{i})\cos[(6x + 8z - 10ct)]$
(C) $\frac{1}{c}(6\hat{k} + 8\hat{i})\cos[(6x + 8z - 10ct)]$ (D) $\frac{1}{c}(6\hat{k} - 8\hat{i})\cos[(6x + 8z + 10ct)]$

PWO083

4. For a plane electromagnetic wave propagating in the $+Z$ direction, which one of the following combination gives the correct possible direction for $\vec{E}$ and $\vec{B}$ field respectively ?

(A) $(\hat{i}+2\hat{j})\ and\ (2\hat{i}-\hat{j})$ (B) $(-2\hat{i}-3\hat{j})\ and\ (3\hat{i}-2\hat{j})$
(C) $(2\hat{i}+3\hat{j})\ and\ (\hat{i}+2\hat{j})$ (D) $(3\hat{i}+4\hat{j})\ and\ (4\hat{i}-3\hat{j})$

PWO252

5. The amplitude of electric field in a parallel light beam of intensity 4 Wm^{-2} is:
(A) 35.5 NC^{-1} (B) 45.5 NC^{-1} (C) 49.5 NC^{-1} (D) 54.8 NC^{-1}

PWO086

6. A plane electromagnetic wave,
$E_z = 100\cos(6\times10^{8}t + 4x)\ V/m$
propagates in a medium of dielectric constant:
(A) 1.5 (B) 2.0 (C) 2.4 (D) 4.0

PWO087

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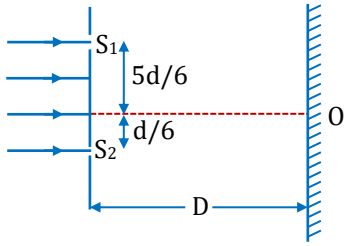
7. The rms value of the electric field of the light coming from the sun is 720 N/c. The average total energy density of the electromagnetic wave is:

(A) $4.58 \times 10^{-6}\ J/m^3$ (B) $6.37 \times 10^{-9}\ J/m^3$ (C) $81.35 \times 10^{-12}\ J/m^3$ (D) $3.3 \times 10^{-3}\ J/m^3$

PWO088

Interference of Light

8. In the figure, if a parallel beam of white light is incident on the plane of the slits S_1 and S_2 then the distance of the nearest white spot on the screen from O is [Assume $D >> d$, $d >> \lambda$]



(A) 0 (B) $\frac{d}{3}$ (C) $\frac{d}{2}$ (D) $\frac{d}{6}$

PWO109

9. At a point on the screen at distance D from the slits in standard YDSE experiment, 3rd maxima is observed at $t = 0$. Now screen is slowly moved with constant speed away from the slits in such a way that the centre of slits and centre of screen always lie on same line and at $t = 1$ sec, the intensity at that point is observed to be $\left(\frac{3}{4}\right)^{th}$ of maximum intensity in between 2nd and 3rd maxima. The speed of screen is:

(A) $\frac{5D}{13}$ (B) $\frac{13D}{5}$ (C) $\frac{17D}{5}$ (D) $\frac{7D}{17}$

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PWO114

10. Light of wavelength λ is incident normally on a thin film. The reflected light shows a minima for reflection. The thickness of film is minimum possible. Now the same film is placed in front of upper slit of the YDSE apparatus. The fringe pattern shifts up by 7.5 mm, the refractive index of material is: (Take: $D = 5\ m$, $d = 0.1\ mm$, $\lambda = 500\ nm$).

(A) 1.25 (B) 2.5 (C) 1.5 (D) 1.75

PWO122

11. What is the minimum thickness of a soap bubble needed for constructive interference in reflected light if the light incident on the film is 900 nm? Refractive index for the film is $\mu = 1.5$.

(A) 100 nm (B) 150 nm (C) 200 nm (D) 250 nm

PWO124

12. To prevent reflection from a camera lens, a thin film of magnesium is coated. In order that the light of wavelength 5000 Å not be reflected the minimum thickness of the film should be (μ for magnesium = 1.25 and μ for glass = 1.5)

(A) $2 \times 10^{-7} m$ (B) $10^{-7}\ m$ (C) $2 \times 10^{-6}\ m$ (D) $2 \times 10^{-8}\ m$

PWO126

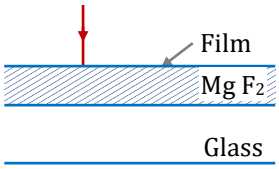
13. White light is incident normally on a thin film which has $n = 1.5$ and a thickness of 5000 Å. For what wavelengths in the visible spectrum (4000 – 7000 Å) will the intensity of the reflected light be a maximum?

(A) 5000 Å, 6000 Å (B) 5260 Å, 6000 Å (C) 4285 Å, 6000 Å (D) 4500 Å, 6500 Å

PWO127

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14. White light is incident normally on a glass surface (n = 1.52) that is coated with a film of MgF_2 (n = 1.38). For what minimum thickness of the film will yellow light of wavelength 550 nm (in air) be missing in the reflected light –



(A) 99.6 nm (B) 49.8 nm (C) 19.6 nm (D) 10.6 nm

PWO128

15. Dipping the frame in a soap solution forms a rectangle film of length ℓ = 0.02 m and height h = 0.030 m. White light falls on the film at an angle α = 30° (measured with respect to the normal direction). The reflected light displays green colour of wavelength λ_0 (500 nm). If density of soap = 1000 kg m^{-3}, then find the mass of the thinnest film? (μ_{soap} = 1.33)

(A) 0.18 mg (B) 0.30 mg (C) 0.06 mg (D) 0.01 mg

PWO129

16. light of wavelength 5880Å is incident on a thin glass plate (μ = 1.5) such that the angle of refraction in the plate is 30°. The minimum thickness of the plate, so that it appears dark in the reflected light will be -

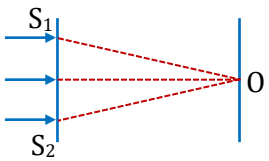
(A) 2940 Å (B) 4074 Å (C) 2263 Å (D) 3394 Å

PWO130

17. **Statement-1:** In YDSE, as shown in figure, central bright fringe is formed at O. If a liquid is filled between plane of slits and screen, the central bright fringe is shifted in upward direction.

and

Statement-2: If path difference at O increases, y-coordinate of central bright fringe will change.



(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

PWO094

18. In a double slit experiment, when the width of one slit is made twice as wide as the other in compared to normal YDSE having slits of equal width. Then, in the interference pattern

(A) the intensities of both the maxima and the minima increase.

(B) the intensity of the maxima increases, and the minima has zero intensity.

(C) the intensity of the maxima decreases and that of the minima increases.

(D) the intensity of the maxima decreases, and the minima has zero intensity.

PWO096

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19. **Statement-1:** In YDSE, the spacing between any two successive points having intensity half of the maximum intensity is same.

and

Statement-2: The intensity on the screen in YDSE varies uniformly with distance from central maximum.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

PWO097

20. In a Young's double slit experiment D equals the distance of screen and d is the separation between the slit. The distance of the nearest point to the central maximum where the intensity is same as that due to a single slit, is equal to:

(A) $\frac{D\lambda}{d}$ (B) $\frac{D\lambda}{2d}$ (C) $\frac{D\lambda}{3d}$ (D) $\frac{2D\lambda}{d}$

PWO101

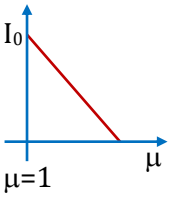
21. In a double slit experiment, the separation between the slits is $d = 0.25\ cm$ and the distance of the screen $D = 100\ cm$ from the slits. If the wavelength of light used is $\lambda = 6000$ Å and I_0 is the intensity of the central bright fringe, the intensity at a distance $x = 4 \times 10^{-5}\ m$ from the central maxima is :-

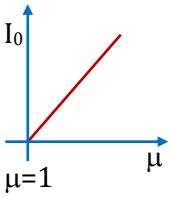
(A) I_0 (B) $\frac{I_0}{2}$ (C) $\frac{3I_0}{4}$ (D) $\frac{I_0}{3}$

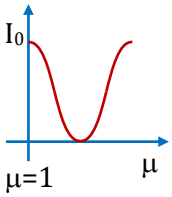
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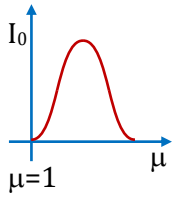
PWO102

22. In a YDSE experiment if a slab whose refractive index can be varied is placed in front of one of the slits then the variation of resultant intensity at mid-point of screen with μ ($\mu \geq 1$) will be best represented by [Assume slits of equal width and there is no absorption by slab]

(A) 

(B) 

(C) 

(D) 

PWO104

23. In a Young's double-slit experiment, let A and B be the two slits. Films of thicknesses t_A and t_B and refractive indices μ_A and μ_B, are placed in front of A and B respectively. If $\mu_A t_A = \mu_B t_B$, the central maximum will:

(A) not shift

(B) shift towards A

(C) shift towards B

(D) option (B), if $t_B > t_A$; option (C) if $t_B < t_A$

PWO105

24. Two monochromatic and coherent point sources of light are placed at a certain distance from each other in the horizontal plane. The locus of all those points in the horizontal plane which have constructive interference will be

(A) a hyperbola

(B) family of hyperbolas

(C) family of straight lines

(D) family of parabolas

PWO107

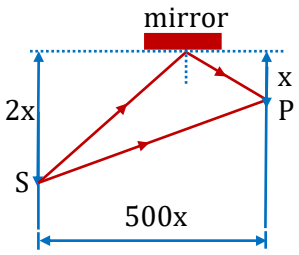
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25. In a YDSE bi-chromatic light of wavelengths 400 *nm* and 560 *nm* are used. The distance between the slits is 0.1 *mm* and the distance between the plane of the slits and the screen is 1 *m*. The minimum distance between two successive regions of complete darkness is :-

(A) 4 *mm* (B) 5.6 *mm* (C) 14 *mm* (D) 28 *mm*

PWO108

26. In the figure shown the point *S* is monochromatic source of light emitting light of wavelength λ. At the point *P* at a distance *x* from the mirror as shown in the figure, interference takes place between two light rays one directly coming from source *S* and another after reflection from the mirror such that a maxima is formed. The minimum value of *x* is

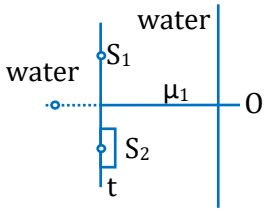


(A) 120 λ (B) 125λ (C) 62.5λ (D) None of these

PWO154

27. A young's double slit experiment is conducted in water (μ_1) as shown in the figure, and a glass plate of thickness *t* and refractive index μ_2 is placed in the path of S_2. Wavelength of light in water is λ. Find the magnitude of the phase difference between waves coming from S_1 and S_2 at '*O*'.

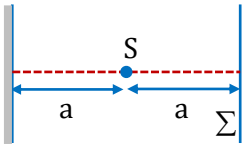
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(A) $\left|\left(\frac{\mu_2}{\mu_1}-1\right)t\right|\frac{2\pi}{\lambda}$ (B) $\left|\left(\frac{\mu_1}{\mu_2}-1\right)t\right|\frac{2\pi}{\lambda}$ (C) $|(\mu_2-\mu_1)t|\frac{2\pi}{\lambda}$ (D) $|(\mu_2-1)t|\frac{2\pi}{\lambda}$

PWO155

28. A point source of light *S* is place in front of a perfectly reflecting mirror as shown in the figure. Σ is a screen. The intensity at the centre of screen is found to be *I*. If the mirror is removed, then the intensity at the centre of screen would be :-

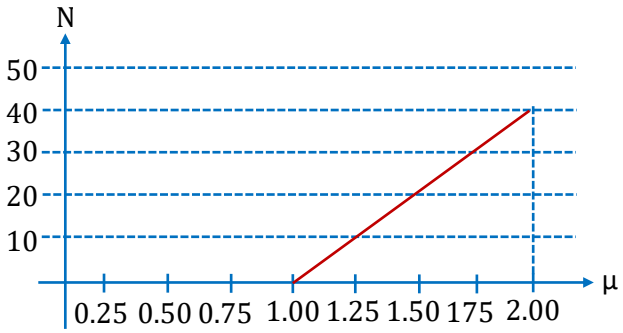


(A) *I* (B) $\frac{10I}{9}$ (C) $\frac{9I}{10}$ (D) $2I$

PWO156

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29. The slits in a double-slit interference experiment are illuminated by orange lights (λ = 600 nm). A thin transparent plastic of thickness t is placed in front of one of the slits. The number of fringes (N) shifting on screen is plotted versus the refractive index μ of the plastic in in graph shown. The value of t is:-

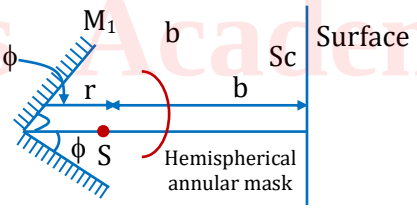


(A) 4.8 μm (B) 48 μm (C) 2.4 μm (D) 24 μm

PWO157

30. Two mirrors M_1 and M_2 make an angle ϕ with line AB. A point source S is kept at a distance r from the point of intersection of mirrors. A small hemispherical annular mask is kept close to S so that no ray emanating from S directly reaches the screen kept at a distance b from the source. Interference pattern of rays reflected from M_1 and M_2 is observed on the screen. Find the fringe width of this interference pattern. λ = wavelength of light being emitted by source.

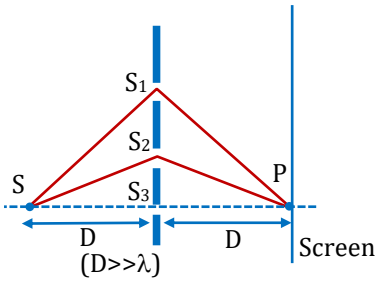
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(A) $\frac{\lambda(b+2r\sin^2\phi)}{2r\sin 2\phi}$ (B) $\frac{\lambda(b+2r\cos 2\phi)}{2r\sin 2\phi}$ (C) $\frac{\lambda(b+r\cos 2\phi)}{2r\sin 2\phi}$ (D) $\frac{\lambda(b+2r\cos 2\phi)}{r\sin 2\phi}$

PWO158

31. A monochromatic light source of wavelength λ is placed at S. Three slits S_1, S_2 and S_3, source S and point P is shown in the figure. If $S_1P - S_2P = \lambda/6$ and $S_1P - S_3P = 2\lambda/3$ and I be the intensity at P when only one slit is open, the intensity at P when all the three slits are open is :-

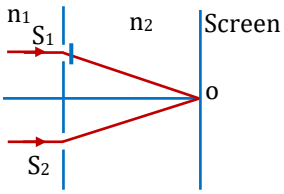


(A) 3I (B) 5I (C) 8I (D) 0

PWO159

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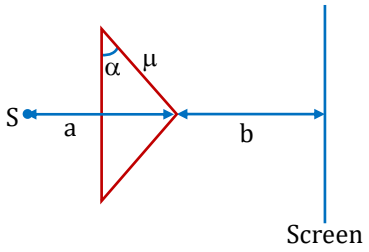
32. In a Young's double slit experiment the slits are illuminated by a parallel beam of light from the medium of refractive index $n_1 = 1.2$. A thin transparent film of thickness $1.2\mu m$ and refractive index $n = 1.5$ is placed infront of s_1 perpendicular to path of light. Wavelength of light measured in medium n_1 is 400 nm. The refractive index of medium between plane of slits & screen is $n_2 = 1.4$. If the light coming from the film and s_2 have equal intensities I then intensity at geometrical centre of the screen is



(A) 0 (B) $2I$ (C) $4I$ (D) None of these

PWO160

33. A fresnel biprism has a slit S as shown. This arrangement is used to obtain interference fringes on the screen. Which of the following modifications would definitely increase the number of fringes obtained on the screen.



(1) Increase a by moving slit further away.
(2) decrease a & increase b by moving biprism towards slit S.
(3) decrease wavelength λ.
(4) decrease angle of prism (α).

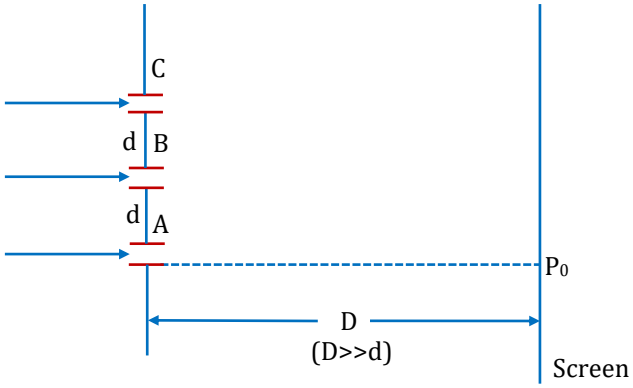
(A) All (B) 1, 2, 3 (C) 1, 2, 4 (D) 1, 3

PWO163

34. Consider superposition of waves coming from three light sources (slits) A, B and C as shown in fig. Given that

$$BP_0 - AP_0 = \frac{\lambda}{3} \quad \text{and} \quad d = \sqrt{\frac{2\lambda D}{3}}$$

The ratio of intensity at P_0 compared to intensity due to individual slit is

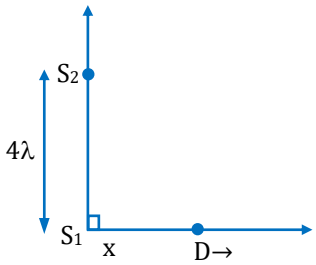


(A) 1.5 (B) 2 (C) 2.5 (D) 3

PWO164

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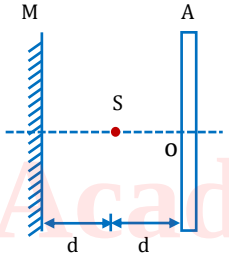
35. Two microwave sources S_1 and S_2 are located at a distance of 4λ from each other along Y-axis, with S_1 at the origin as shown in the figure. A microwave detector D moves along the X-axis away from S_1. The first maximum is obtained for x (> 0) equal to: (wavelength of the microwave is λ):



(A) λ (B) $\frac{1}{6}\lambda$ (C) 3λ (D) $\frac{7}{6}\lambda$

PWO165

36. A point source of light 'S' at a distance d from the screen A, produces light intensity I_0 at the centre O of the screen. If a completely reflecting mirror M is placed at a distance d behind the source as shown in the figure, treating object and its image as incoherent with each other, find the intensity at the centre of the screen.



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(A) $\frac{9}{10}I_0$ (B) $\frac{10}{9}I_0$ (C) $\frac{8}{9}I_0$ (D) $\frac{9}{8}I_0$

PWO166

Diffraction

37. Consider Fraunhofer diffraction pattern obtained with a single slit illuminated at normal incidence. At the angular position of the first diffraction minimum, the phase difference (in radians) between the wavelets from the opposite edges of the slit is:

(A) π (B) 2π (C) $\pi/4$ (D) $\pi/2$

PWO131

38. The image of a star (effectively a point source) is made by a convergent lens of focal length 1m and diameter of aperture 5.0 cm. If the lens is ideal and the effective wavelength in image formation is taken as 5×10^{-5} cm, the diameter of the image formed will be nearest to:

(A) zero (B) 10^{-6} cm (C) 10^{-5} cm (D) 10^{-3} cm

PWO136

39. A microscope objective gathers light over a cone of semi-vertex 30° and used visible light of wavelength 5500 Å. Its resolving limit is :

(A) 2.5×10^{-5} cm (B) 4.6×10^{-5} cm (C) 3.9×10^{-5} cm (D) 6.7×10^{-5} cm

PWO137

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40. A person wishes to distinguish between two pillars located at a distance of 11 *km*. what should be the minimum distance between these pillars? [*R.L.* = 1'(one minute)]

(A) 6.4 (B) 1.6 (C) 3.2 (D) None of these

PWO138

41. Find limit of resolution of a microscope if an object of numerical aperture 0.06 is viewed by using light of wavelength 9600 Å.

(A) 2.69 (B) 3.59 (C) 6.09 (D) 9.76

PWO140

42. Find ratio of resolving powers of an optical microscope for two wave lengths λ_1 = 4200 Å and λ_2 = 7200 Å.

(A) $\frac{7}{12}$ (B) $\frac{12}{7}$ (C) $\frac{6}{7}$ (D) $\frac{6}{14}$

PWO141

43. Two towers on top of two hills are 40 *km* apart. The line joining them passes 50 *m* above a hill halfway the towers. What is the longest wavelength of radio waves, which can be sent between the towers without appreciable diffraction effects?

(A) 10.5 *cm* (B) 11.5 *cm* (C) 12.5 *cm* (D) 13.5 *cm*

PWO142

44. Consider diffraction by a slit of small width. *O* is point of central maxima; *A* is point closest to *O* where intensity is zero. Now if width of slit is increased slightly.

(A) Intensity of *O* increases and point *A* shifts away from *O*.

(B) Intensity of *O* increases and point *A* shifts towards *O*.

(C) Intensity of *O* decreases and point *A* shifts aways from *O*.

(D) Intensity of *O* decreases and point *A* shifts towards *O*.

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PWO167

45. In single slit fraunhoffer diffraction pattern, resultant amplitude at the centre of central maxima is A_0 and it is A at an angular position of 30°. Then $\frac{A}{A_0}$ will be - [Slit width $a = \lambda$]

(A) $\frac{\pi}{2}$ (B) $\frac{2}{\pi}$ (C) $\frac{\pi}{4}$ (D) $\frac{4}{\pi}$

PWO168

46. What is the wavelength of a beam of *x* rays which is diffracted by a crystal in a direction making an angle of 60° with the Incident beam and corresponding to first order 'reflection' from crystal lattice planes separated by 3×10^{-10} *m*?

(A) 6 Å (B) 9 Å (C) 3 Å (D) 4 Å

PWO169

47. If I_0 is the intensity of the principal maxima in a Fraunhoffer single slit diffraction pattern, then what will be the intensity of the first secondary maxima :-

(A) $\frac{2}{5\pi^2} I_0$ (B) $\frac{1}{9\pi} I_0$ (C) $\frac{3}{5\pi^2} I_0$ (D) $\frac{4}{9\pi^2} I_0$

PWO170

For More Join: @IITJEE_Advanced

Polarisation

48. Light waves can be polarised as they are

(A) Transverse (B) of high frequency (C) Longitudinal (D) Reflected

PWO143

49. Plane polarised light is passed through a polaroid. On viewing through the polaroid, we find that when the polaroid is given one complete rotation about the direction of the light, one of the following is observed.

(A) The intensity of light gradually decreases to zero and remains at zero.

(B) The intensity of light gradually increases to a maximum and remains at maximum.

(C) There is no change in intensity.

(D) The intensity of light is twice maximum and twice zero.

PWO146

50. A ray of light is incident on the surface of a glass plate at an angle of incidence equal to Brewster's angle ϕ. If μ represents the refractive index of glass with respect to air, then the angle between reflected and refracted rays is

(A) $90 + \phi$ (B) $\sin^{-1}(\mu \cos\phi)$ (C) $90°$ (D) $90^{o} - \sin^{-1}(\sin\phi/\mu)$

PWO147

51. A light has amplitude A and angle between analyser and polariser is 60°. Light is reflected by analyser has amplitude.

(A) $A\sqrt{2}$ (B) $A/\sqrt{2}$ (C) $\sqrt{3}A/2$ (D) $A/2$

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PWO148

52. Unpolarized light of intensity 32 Wm^{-2} passes through three polarizers such that transmission axis of the first and second polarizer makes an angle of 30° with each other and the transmission axis of the last polarizer is crossed with that of the first. The intensity of final emerging light will be

(A) 32 Wm^{-2} (B) 3 Wm^{-2} (C) 8 Wm^{-2} (D) 4 Wm^{-2}

PWO149

53. Two polaroid are placed in the path of unpolarized beam of intensity I_0 such that no light is emitted from the second polaroid. If a third polaroid whose polarization axis makes an angle θ with the polarization axis of first polaroid, is placed between these polaroids then the intensity of light emerging from the last polaroid will be

(A) $\left(\frac{I_0}{8}\right)\sin^2 2\theta$ (B) $\left(\frac{I_0}{4}\right)\sin^2 2\theta$ (C) $\left(\frac{I_0}{2}\right)\cos^4\theta$ (D) $I_0\cos^4\theta$

PWO150

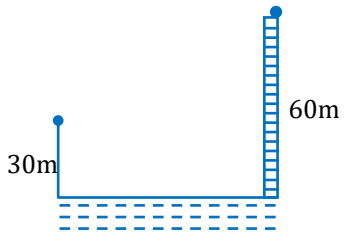
54. When the angle of incidence on a material is 60°, the reflected light is completely polarized. The velocity of the refracted ray inside the material is (in ms^{-1})

(A) 3×10^8 (B) $\left(\frac{3}{\sqrt{2}}\right) \times 10^8$ (C) $\sqrt{3} \times 10^8$ (D) 0.5×10^8

PWO152

For More Join: @IITJEE_Advanced

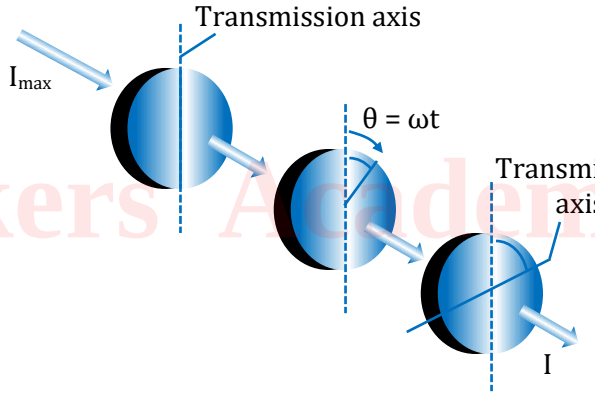
55. A person lives in a high rise building at a height $30m$ on the bank of a river. Across the river is a well it towers of height $60m$. When person looks through a polarizer at an appropriate angle at light from top of tower reflecting from the river surface, he finds intensity of light is least. If refractive index of water is $\frac{4}{3}$, width of river is:-



(A) $120m$ (B) $80m$ (C) $60m$ (D) $40m$

PWO172

56. In figure suppose the transmission axes of the left and right polarizing disks are perpendicular to each other. Also, let the center disk be rotated on the common axis with an angular speed ω. That if unpolarized light is incident on the left disk with an intensity I_{max} the intensity of the beam emerging from the right disk is given by :-



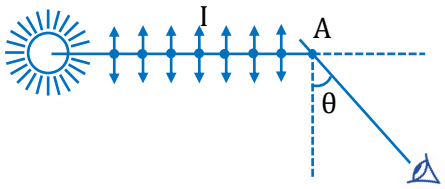
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(A) $I = \frac{1}{16} I_{max}(1 - \cos 4\omega t)$ (B) $I = \frac{1}{16} I_{max}(1 - \cos 2\omega t)$

(C) $I = \frac{1}{8} I_{max}(1 - \cos 4\omega t)$ (D) $I = \frac{1}{8} I_{max}(1 - \cos 2\omega t)$

PWO173

57. Unpolarized light of intensity I scattered from point A. Intensity of light perceive by given observer is :-



(A) I (B) $\frac{I}{2}\cos^2\theta$ (C) $\frac{I}{2} + \frac{I}{2}\sin^2\theta$ (D) $\frac{I}{2} + \frac{I}{2}\cos^2\theta$

PWO174

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EXERCISE (O-2)

MULTIPLE CORRECT TYPE QUESTIONS

1. As a wave propagates,

(A) the wave intensity remains constant for a plane wave.

(B) the wave intensity decreases with distance from source and is proportional to inverse of the distance, for a spherical wave.

(C) the wave intensity decreases with distance from source and proportional to inverse square of the distance, for a spherical wave.

(D) total average power of the spherical wave over any spherical surface centered at the source remains same.

PWO092

2. To observe a sustained interference pattern formed by two light waves, it is not necessary that they must have:

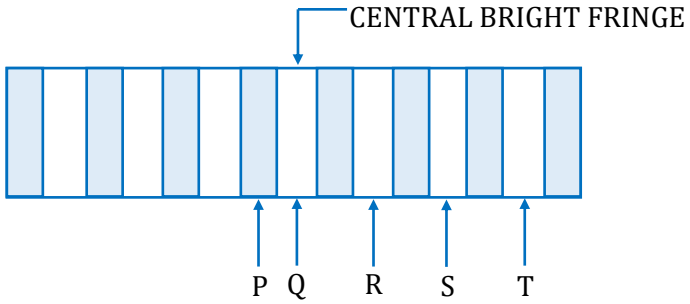
(A) the same frequency (B) same amplitude

(C) a constant phase difference (D) the same intensity

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PWO095

3. The figure shows the interference pattern obtained in double-slit experiment using light of wavelength 600 *nm*. *P*, *Q*, *R*, *S* and *T* are marked on five fringes.



(A) The third order bright fringe is *R*.

(B) The third order bright fringe is *T*.

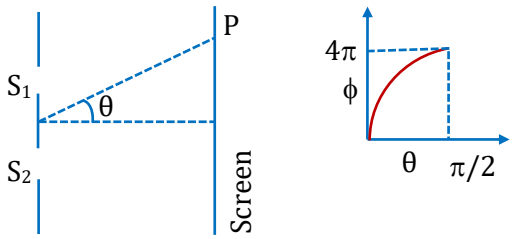
(C) The fringe resulting from a phase difference of 4π is *S*.

(D) If ΔX_A and ΔX_C represent path differences between waves interfering at *P* and *R* respectively then $(|\Delta X_C| - |\Delta X_A|)$ is equal to 300*nm*.

PWO175

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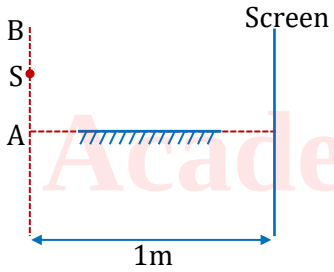
4. In a YDSE setup, we plot the phase difference (ϕ) between both waves at point P on the screen against the angular position (θ) of point P on the screen. The graph is as shown below.



(A) The distance $S_1S_2 = 2\lambda$
(B) There are a total of 4 minima on the screen
(C) The first maxima above the centre is at $\theta = \frac{\pi}{4}$
(D) At $\theta = \frac{\pi}{6}$, intensity is maximum

PWO176

5. In an interference experiment as shown in the figure, the source plane and screen are separated by a distance $1m$. At a certain position of source, fringe width is $\frac{1}{4}mm$ and by moving the source away from mirror along the line AB by 0.6 mm, the fringe width changed to $\frac{1}{6}mm$:-

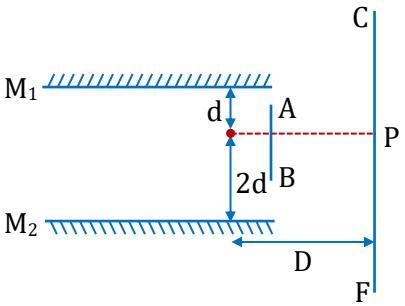


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(A) Wavelength of light used is 5000 Å
(B) Wavelength of light used is 6000 Å
(C) Initial distance of source from A is 1.2 mm
(D) Initial distance of source from A is 0.6 mm

PWO120

6. M_1 and M_2 are two plane mirror and S is monochromatic source. AB is a 'stop' which stops direct light to reach CF and allows reflected light from M_1 and M_2 to reach CF. P is a point on CF such that SP is parallel to M_1 and M_2 and perpendicular to CF :



(A) Circular fringes will be formed on CF
(B) P will be central maxima
(C) P will be a point of maxima if $\lambda = \frac{6d^2}{D}$
(D) P will be a point of maxima if $\lambda = \frac{d^2}{D}$

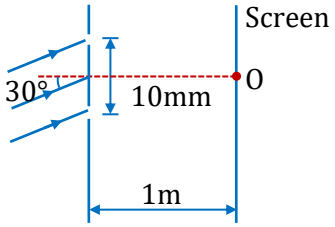
PWO121

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COMPREHENSION TYPE QUESTIONS

PARAGRAPH FOR QUESTION NO. 7 TO 9

In a Young's double slit experiment, a parallel beam of light of wavelength 4000 $\mathring{A}$ is incident on the slit at an angle of 30° as shown in the figure.



7. Find fringe width of the interference pattern

(A) 40 μm (B) 20 μm (C) 30 μm (D) 10 μm

PWO111

8. Find phase difference between the light coming from the two slits at O.

(A) zero (B) $\pi/2$ (C) $\pi/6$ (D) $\pi/4$

PWO112

9. Find the position of centre bright fringe on screen.

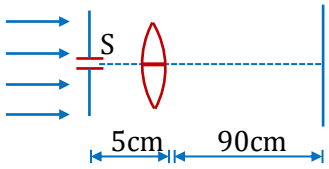
(A) 50 cm (B) 30 cm (C) 40 cm (D) 45 cm

PWO113

PARAGRAPH FOR QUESTION NO. 10 TO 12

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A convex lens of focal length f = 10 cm is cut through a optical axis into two semicircular halves. In doing so, glass of 0.5 mm thickness is lost. The two halves are then joined together. This lens is placed symmetrically 5 cm in front of a slit which is illuminated with light of wavelength $\lambda = 5000\mathring{A}$. A screen is placed on the other side of the lens at a distance of 90 cm from the lens, perpendicular to the optical axis of the lens.



10. What is the distance between two images of S ?

(A) $\frac{1}{2}\,mm$ (B) 1 mm (C) $\frac{1}{4}\,mm$ (D) $\frac{3}{2}\,mm$

PWO117

11. What is the fringe width on the screen?

(A) $\frac{1}{2}\,mm$ (B) $\frac{3}{4}\,mm$ (C) 1 mm (D) $\frac{5}{4}\,mm$

PWO118

12. Total number of maximas on the screen are

(A) 5 (B) 4 (C) 2 (D) 7

PWO119

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MATRIX MATCH TYPE QUESTION

13. Match List-I with List-II.

	List-I		**List-II**
(a)	Source of microwave frequency	(i)	Radioactive decay on nucleus
(b)	Source of infrared frequency	(ii)	Magnetron
(c)	Source of Gamma Rays	(iii)	Inner shell electrons
(d)	Source of *X*-rays	(iv)	Vibration of atoms and molecules
		(v)	LASER
		(vi)	*RC* circuit

PWO080

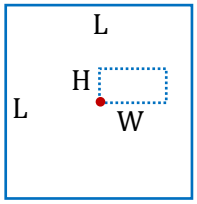
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EXERCISE (S)

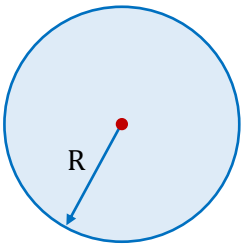
EM Wave

1. A capacitor with square plates of edge length L is being discharged by a current of $0.75A$. Figure is a head-on view of one of the plates from inside the capacitor. A dashed rectangular path is shown. If $L = 12\ cm$, $W = 4.0\ cm$, and $H = 2.0\ cm$, what is the value of $\int \vec{B}\cdot d\vec{s}$ around the dashed path?



PWO200

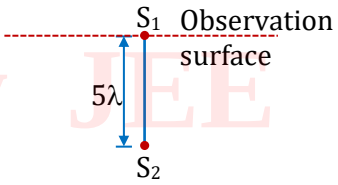
2. Nonuniform electric flux figure shows a circular region of radius $R = 3.00\ cm$ in which an electric flux is directed out of the plane of the page. The flux encircled by a concentric circle of radius r is given by $\phi_{E,\,enc} = (0.600\ V.\ m/s)\ (r/R)t$, where $r \le R$ and t is in seconds. What is the magnitude of the induced magnetic field at radial distances (a) 2.00 cm and (b) 5.00 cm?



PWO201

Interference of Light

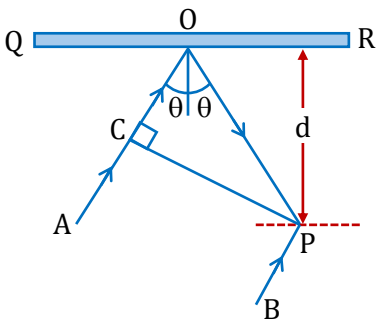
3. Two microwave coherent point sources emitting waves of wavelength λ are placed at 5λ distance apart. The interference is being observed on a flat non-reflecting surface along a line passing through one source, in a direction perpendicular to the line joining the two sources (refer figure). Considering λ as 4 mm, calculate the positions of maxima and draw shape of interference pattern. Take initial phase difference between the two sources to be zero.



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PWO115

4. In the given diagram, CP represents a wavefront and AO and BP, the corresponding two rays. The condition on q for constructive interference at P between the ray BP and reflected ray OP is $\cos\theta = \lambda/nd$. Find n where n is an integer



5. Two coherent waves are described by the following expressions.

$$E_1 = E_0 \sin\left(\frac{2\pi x_1}{\lambda} - 2\pi f t + \frac{\pi}{6}\right)\ ;\ E_2 = E_0 \sin\left(\frac{2\pi x_2}{\lambda} - 2\pi f t + \frac{\pi}{8}\right)$$

Determine the relationship between x_1 and x_2 that produces constructive interference when the two waves are superposed.

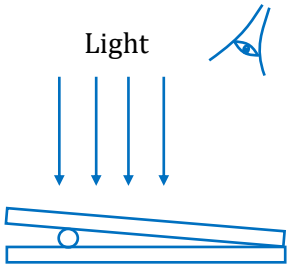
PWO099

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6. One slit of a double slit experiment is covered by a thin glass plate of refractive index 1.4 and the other by a thin glass plate of refractive index 1.7. The point on the screen, where central bright fringe was formed before the introduction of the glass sheets, is now occupied by the 5th bright fringe. Assuming that both the glass plates have same thickness and wavelength of light used is 4800 $\mathring{A}$, find their thickness.

PWO106

7. A very fine hair-like fiber is stuck between two microscope slides. As a result of this, there is a wedge of air between them. When the slides are illuminated normally (from above) with light of wavelength 550 *nm*, bright and dark interference bands are formed. The fiber is seen to lie at the position of fifth dark band counting from the common edge. What is the diameter (in *nm*) of the fiber?



PWO183

8. A narrow light beam is incident on a plane-parallel plate having a refractive index of $n = \frac{17}{16}$ at an angle of 30° to the normal from air. As a result, it is partially reflected and refracted. The refracted beam is reflected by the rear surface of the plate and then undergoes refraction again, emerging from the plate with a displacement of $4\sqrt{3}$ *cm* parallel to the primary reflected beam. The thickness of the plate is given as '*t*' *m*. Find the value of 40 *t* .

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PWO184

Diffraction

9. A horizontal laser beam of wavelength 660 *nm* has a circular cross section 2.00 *mm* in diameter. A rectangular aperture is to be placed in the center of the beam so that when the light falls perpendicularly on a wall 4.50 *m* away, the central maximum fills a rectangle 110 *mm* wide and 6.00 *mm* high. The dimensions are measured between the minima bracketing the central maximum.

(a) Find the required width and height of the aperture.

(b) Is the longer dimension of the aperture horizontal or vertical ?

PWO133

Polarisation

10. A beam of plane-polarized light falls on a polarized which rotates about the axis of the ray with angular velocity $\omega = 21$ *rad/s*. Find the energy of light passing through the polarizer per one revolution if the flux of energy of the incident ray is equal to $\varphi_0 = 4.0$ *mW*.

PWO185

11. A beam of natural light galls on a system on $N = 6$ Nicol prisms whose transmission planes are turned each through an angle $\phi = 30°$ with respect to that of the foregoing prism. What fraction of luminous flux passes through this system?

PWO186

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EXERCISE - JEE (Main) PYQ

1. The box of a pin hole camera, of length L, has a hole of radius a. It is assumed that when the hole is illuminated by a parallel beam of light of wavelength λ the spread of the spot (obtained on the opposite wall of the camera) is the sum of its geometrical spread and the spread due to diffraction. The spot would then have its minimum size (say b_{min}) when :- **[JEE (Main) 2016]**

(1) $a = \frac{\lambda^2}{L}$ and $b_{min} = \sqrt{4\lambda L}$ (2) $a = \frac{\lambda^2}{L}$ and $b_{min} = \left(\frac{2\lambda^2}{L}\right)$

(3) $a = \sqrt{\lambda L}$ and $b_{min} = \left(\frac{2\lambda^2}{L}\right)$ (4) $a = \sqrt{\lambda L}$ and $b_{min} = \sqrt{4\lambda L}$

PWO037

2. An EM wave from air enters a medium. The electric fields are $\vec{E}_1 = E_{01}\hat{x}\cos\left[2\pi\nu\left(\frac{z}{c} - t\right)\right]$in air and $\vec{E}_2 = E_{02}\hat{x}\cos[k(2z - ct)]$in medium, where the wave number k and frequency ν refer to their values in air. The medium is non-magnetic. If $\in_{r_1}$ and $\in_{r_2}$ refer to relative permittivity of air and medium respectively, which of the following options is correct ? **[JEE (Main) 2018]**

(1) $\frac{\in_{r_1}}{\in_{r_2}} = 2$ (2) $\frac{\in_{r_1}}{\in_{r_2}} = \frac{1}{4}$ (3) $\frac{\in_{r_1}}{\in_{r_2}} = \frac{1}{2}$ (4) $\frac{\in_{r_1}}{\in_{r_2}} = 4$

PWO039

3. A light wave is incident normally on a glass slab of refractive index 1.5. If 4% of light gets reflected and the amplitude of the electric field of the incident light is $30 V/m$, then the amplitude of the electric field for the wave propagating in the glass medium will be: **[JEE (Main) 2019]**

(1) 10 V/m (2) 24 V/m (3) 30 V/m (4) 6 V/m

PWO043

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4. The electric field of a plane electromagnetic wave is given by $\vec{E} = E_0\frac{\hat{i}+\hat{j}}{\sqrt{2}}\cos(kz + \omega t)$

At $t = 0$, a positively charged particle is at the point $(x, y, z) = \left(0,0,\frac{\pi}{k}\right)$. If its instantaneous velocity at $(t = 0)$ is $v_0\,\hat{k}$, the force acting on it due to the wave is: **[JEE (Main) 2020]**

(1) zero (2) parallel to $\frac{\hat{i}+\hat{j}}{\sqrt{2}}$ (3) antiparallel to $\frac{\hat{i}+\hat{j}}{\sqrt{2}}$ (4) parallel to $\hat{k}$

PWO045

5. In a plane electromagnetic wave, the directions of electric field and magnetic field are represented by $\hat{k}$ and $2\hat{i} - 2\hat{j}$, respectively. What is the unit vector along direction of propagation of the wave. **[JEE (Main) 2020]**

(1) $\frac{1}{\sqrt{2}}(\hat{i} + \hat{j})$ (2) $\frac{1}{\sqrt{5}}(\hat{i} + 2\hat{j})$ (3) $\frac{1}{\sqrt{5}}(2\hat{i} + \hat{j})$ (4) $\frac{1}{\sqrt{2}}(\hat{j} + \hat{k})$

PWO046

6. The magnetic field vector of an electromagnetic wave is given by $B = B_0\frac{\hat{i}+\hat{j}}{\sqrt{2}}\cos(kz - \omega t)$; where $\hat{i}, \hat{j}$ represents unit vector along x and y-axis respectively. At $t = 0s$, two electric charges q_1 of 4π coulomb and q_2 of 2π coulomb located at $\left(0,0,\frac{\pi}{k}\right)$ and $\left(0,0,\frac{3\pi}{k}\right)$, respectively, have the same velocity of $0.5\,c\hat{i}$ (where c is the velocity of light). The ratio of the force acting on charge q_1 to q_2 is: **[JEE (Main) 2021]**

(1) $2\sqrt{2} : 1$ (2) $1 : \sqrt{2}$ (3) $2 : 1$ (4) $\sqrt{2} : 1$

PWO047

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7. Electric field of plane electromagnetic wave propagating through a non-magnetic medium is given by $E = 20\cos(2 \times 10^{10}\, t - 200x)\ V/m$. The dielectric constant of the medium is equal to:
(Take $\mu_r = 1$) **[JEE (Main) 2021]**

(1) 9 (2) 2 (3) $\frac{1}{3}$ (4) 3

PWO048

8. Light wave travelling in air along x-direction is given by $E_y = 540 \sin \pi \times 10^4(x - ct)\ Vm^{-1}$. Then, the peak value of magnetic field of wave will be (Given $c = 3 \times 10^8\ ms^{-1}$) **[JEE (Main) 2022]**

(1) $18 \times 10^{-7}T$ (2) $54 \times 10^{-7}T$ (3) $54 \times 10^{-8}T$ (4) $18 \times 10^{-8}T$

PWO049

9. The ratio of average electric energy density and total average energy density of electromagnetic wave is: **[JEE (Main) 2023]**

(1) 2 (2) 1 (3) 3 (4) $\frac{1}{2}$

PWO050

10. A beam of light consisting of two wavelengths 7000 Å and 5500 Å is used to obtain interference pattern in Young's double slit experiment. The distance between the slits is 2.5 mm and the distance between the plane of slits and the screen is 150 cm. The least distance from the central fringe, where the bright fringes due to both the wavelengths coincide, is $n \times 10^{-5}\ m$. The value of n is_____. **[JEE (Main) 2023]**

PWO187

11. A monochromatic light wave with wavelength λ_1 and frequency ν_1 in air enters another medium. If the angle of incidence and angle of refraction at the interface are 45° and 30° respectively, then the wavelength λ_2 and frequency ν_2 of the refracted wave are : **[JEE (Main) 2023]**

(1) $\lambda_2 = \lambda_1, \nu_2 = \sqrt{2}\nu_1$ (2) $\lambda_2 = \frac{1}{\sqrt{2}}\lambda_1, \nu_2 = \nu_1$

(3) $\lambda_2 = \sqrt{2}\lambda_1, \nu_2 = \nu_1$ (4) $\lambda_2 = \lambda_1, \nu_2 = \frac{1}{\sqrt{2}}\nu_1$

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PWO188

12. The width of fringe is 2 mm on the screen in a double slits experiment for the light of wavelength of 400 nm. The width of the fringe for the light of wavelength 600 nm will be: **[JEE (Main) 2023]**

(1) 4 mm (2) 1.33 mm (3) 3 mm (4) 2 mm

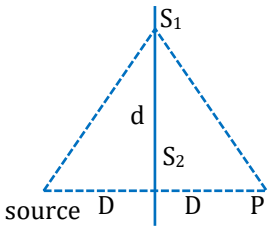
PWO189

13. A single slit of width a is illuminated by a monochromatic light of wavelength 600 nm. The value of 'a' for which first minimum appears at $\theta = 30°$ on the screen will be : **[JEE (Main) 2023]**

(1) 0.6 μm (2) 1.2 μm (3) 1.8 μm (4) 3 μm

PWO190

14. In a double slit experiment shown in figure, when light of wavelength 400 nm is used, dark fringe is observed at P. If $D = 0.2\ m$. the minimum distance between the slits S_1 and S_2 is ____ mm.
[JEE (Main) 2024]



PWO202

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15. In Young's double slit experiment, light from two identical sources are superimposing on a screen. The path difference between the two lights reaching at a point on the screen is $\frac{7\lambda}{4}$. The ratio of intensity of fringe at this point with respect to the maximum intensity of the fringe is :

[JEE (Main) 2024]

(1) 1/2 (2) 3/4 (3) 1/3 (4) 1/4

PWO203

16. In Young's double slit experiment, carried out with light of wavelength 5000Å, the distance between the slits is 0.3 *mm* and the screen is at 200 *cm* from the slits. The central maximum is at $x = 0$ *cm*. The value of x for third maxima is _____*mm*.

[JEE (Main) 2024]

PWO204

17. Match List I with List II

List-I		List-II	
A.	$\oint \vec{B}.\vec{dl} = \mu_0 i_c + \mu_0 \varepsilon_0 \frac{d\phi_E}{dt}$	I.	Gauss' law for electricity
B.	$\oint \vec{E}.\vec{dl} = \frac{d\phi_B}{dt}$	II.	Gauss' law for magnetism
C.	$\oint \vec{E}.\vec{dA} = \frac{Q}{\varepsilon_0}$	III.	Faraday law
D.	$\oint \vec{B}.\vec{dA} = 0$	IV.	Ampere – Maxwell law

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[JEE (Main) 2024]

Chose the correct answer from the options given below

(1) A-IV, B-I, C-III, D-II (2) A-II, B-III, C-I, D-IV

(3) A-IV, B-III, C-I, D-II (4) A-I, B-II, C-III, D-IV

PWO205

18. The magnetic field in a plane electromagnetic wave is $B_y = (3.5 \times 10^{-7}) \sin(1.5 \times 10^3 x + 0.5 \times 10^{11} t) T$. The corresponding electric field will be

[JEE (Main) 2024]

(1) $E_y = 1.17 \sin(1.5 \times 10^3 x + 0.5 \times 10^{11} t) Vm^{-1}$

(2) $E_z = 105 \sin(1.5 \times 10^3 x + 0.5 \times 10^{11} t) Vm^{-1}$

(3) $E_z = 1.17 \sin(1.5 \times 10^3 x + 0.5 \times 10^{11} t) Vm^{-1}$

(4) $E_y = 10.5 \sin(1.5 \times 10^3 x + 0.5 \times 10^{11} t) Vm^{-1}$

PWO206

19. Due to presence of an em-wave whose electric component is given by $E = 100 \sin(\omega t - kx)\ NC^{-1}$, a cylinder of length 200 *cm* holds certain amount of em-energy inside it. If another cylinder of same length but half diameter than previous one holds same amount of em-energy, the magnitude of the electric field of the corresponding em-wave should be modified as

[JEE (Main) 2025]

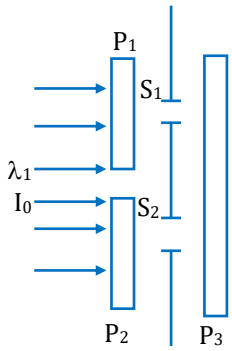
(1) $25 \sin(\omega t - kx)\ NC^{-1}$ (2) $200 \sin(\omega t - kx)\ NC^{-1}$

(3) $400 \sin(\omega t - kx)\ NC^{-1}$ (4) $50 \sin(\omega t - kx)\ NC^{-1}$

PWO250

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20. In a Young's double slit experiment, three polarizers are kept as shown in the figure. The transmission axes of P_1 and P_2 are orthogonal to each other. The polarizer P_3 covers both the slits with its transmission axis at 45° to those of P_1 and P_2. An unpolarized light of wavelength λ and intensity I_0 is incident on P_1 and P_2. The intensity at a point after P_3 where the path difference between the light waves from s_1 and s_2 is $\frac{\lambda}{3}$, is



[JEE (Main) 2025]

(1) $\frac{I_0}{2}$ (2) $\frac{I_0}{4}$ (3) I_0 (4) $\frac{I_0}{3}$

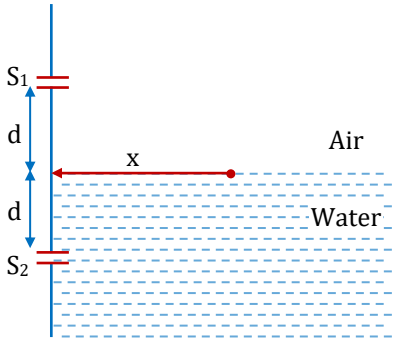
PWO251

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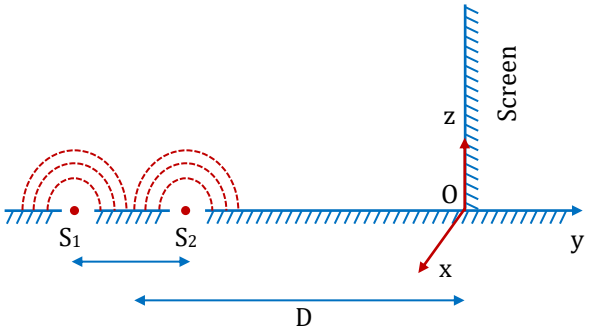
EXERCISE - JEE (Advanced) PYQ

1. A Young's double slit interference arrangement with slits S_1 and S_2 is immersed in water (refractive index = 4/3) as shown in the figure. The positions of maxima on the surface of water are given by $x^2 = p^2m^2\lambda^2 - d^2$, where λ is the wavelength of light in air (refractive index = 1), $2d$ is the separation between the slits and m is an integer. The value of p is. **[JEE (Advanced) 2015]**



PWO073

2. While conducting the Young's double slit experiment, a student replaced the two slits with a large opaque plate in the x-y plane containing two small holes that act as two coherent point sources (S_1, S_2) emitting light of wavelength 600 nm. The student mistakenly placed the screen parallel to the x-z plane (for $z > 0$) at a distance $D = 3m$ from the mid-point of S_1S_2, as shown schematically in the figure. The distance between the sources $d = 0.6003$ mm. The origin O is at the intersection of the screen and the line joining S_1S_2. Which of the following is (are) true of the intensity pattern on the screen ? **[JEE (Advanced) 2016]**



(A) Hyperbolic bright and dark bands with foci symmetrically placed about O in the x-direction

(B) Semi-circular bright and dark bands centered at point O

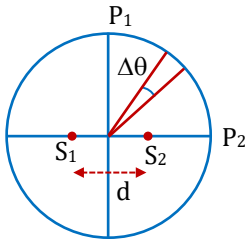
(C) The region very close to the point O will be dark

(D) Straight bright and dark bands parallel to the x-axis

PWO074

For More Join: @IITJEE_Advanced

3. Two coherent monochromatic point sources S_1 and S_2 of wavelength λ = 600 *nm* are placed symmetrically on either side of the center of the circle as shown. The sources are separated by a distance d = 1.8 *mm*. This arrangement produces interference fringes visible as alternate bright and dark spots on the circumference of the circle. The angular separation between two consecutive bright spots is $\Delta\theta$. Which of the following options is/are correct?



[JEE (Advanced) 2017]

(A) A dark spot will be formed at the point P_2.

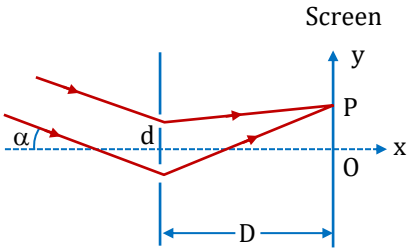
(B) The angular separation between two consecutive bright spots decreases as we move from P_1 to P_2 along the first quadrant.

(C) At P_2 the order of the fringe will be maximum.

(D) The total number of fringes produced between P_1 and P_2 in the first quadrant is close to 3000.

PWO075

4. In a Young's double slit experiment, the slit separation d is 0.3*mm* and the screen distance D is 1*m*. A parallel beam of light of wavelength 600 *nm* is incident on the slits at angle α as shown in figure. On the screen, the point O is equidistant from the slits and distance PO is 11.0 *mm*. Which of the following statement(s) is/are correct?



[JEE (Advanced) 2019]

(A) For $\alpha = \frac{0.36}{\pi}$ degree, there will be destructive interference at point O.

(B) Fringe spacing depends on α

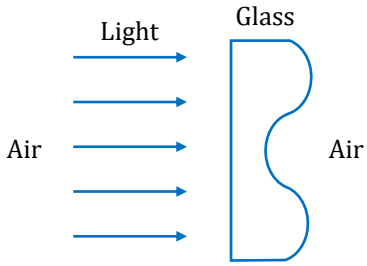
(C) For $\alpha = \frac{0.36}{\pi}$ degree, there will be destructive interference at point P

(D) For α = 0, there will be constructive interference at point P.

PWO076

5. A parallel beam of light strikes a piece of transparent glass having cross section as shown in the figure below. Correct shape of the emergent wavefront will be (figures are schematic and not drawn to scale)-

[JEE (Advanced) 2020]



(A) 

(B) 

(C) 

(D) 

PWO077

For More Join: @IITJEE_Advanced

6. A physical quantity $\vec{S}$ is defined as $\vec{S} = (\vec{E} \times \vec{B}) / \mu_0$, where $\vec{E}$ is electric field, $\vec{B}$ is magnetic field and μ_0 is the permeability of free space. The dimensions of $\vec{S}$ are the same as the dimensions of which of the following quantity (ies)? **[JEE (Advanced) 2021]**

(A) $\frac{\text{Energy}}{\text{Charge} \times \text{current}}$ (B) $\frac{\text{Force}}{\text{Length} \times \text{Time}}$ (C) $\frac{\text{Energy}}{\text{Volume}}$ (D) $\frac{\text{Power}}{\text{Area}}$

PWO078

7. The electric field associated with an electromagnetic wave propagating in a dielectric medium is given by $\vec{E} = 30(2\hat{x} + \hat{y}) \sin\left[2\pi\left(5 \times 10^{14} t - \frac{10^7}{3} z\right)\right] V\ m^{-1}$. Which of the following option(s) is(are) correct?

[Given: The speed of light in vacuum, $c = 3 \times 10^8\ ms^{-1}$] **[JEE (Advanced) 2023]**

(A) $B_x = -2 \times 10^{-7} \sin\left[2\pi\left(5 \times 10^{14} t - \frac{10^7}{3} z\right)\right] Wbm^{-2}$

(B) $B_y = 2 \times 10^{-7} \sin\left[2\pi\left(5 \times 10^{14} t - \frac{10^7}{3} z\right)\right] Wbm^{-2}$

(C) The wave is polarized in the xy-plane with polarization angle 30° with respect to the x-axis.

(D) The refractive index of the medium is 2.

PWO191

8. A point source S emits unpolarized light uniformly in all directions. At two points A and B, the ratio $r = I_A / I_B$ of the intensities of light is 2. If a set of two polaroids having 45° angle between their pass-axes is placed just before point B, then the new value of r will be ____.

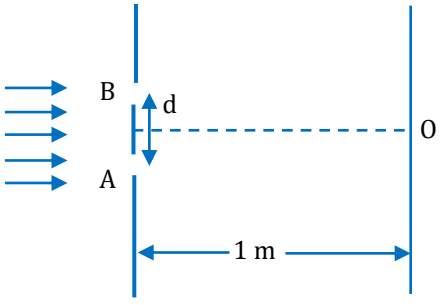
[JEE (Advanced) 2024]

PWO207

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PARAGRAPH FOR QUESTION NO. 9 AND 10

In a Young's double slit experiment, each of the two slits A and B, as shown in the figure, are oscillating about their fixed center and with a mean separation of 0.8 mm. The distance between the slits at time t is given by $d = (0.8 + 0.04 \sin \omega t)$ mm, where $\omega = 0.08\ rad\ s^{-1}$. The distance of the screen from the slits is 1 m and the wavelength of the light used to illuminate the slits is 6000 Å. The interference pattern on the screen changes with time, while the central bright fringe (zeroth fringe) remains fixed at point O.



[JEE (Advanced) 2024]

9. The 8th bright fringe above the point O oscillates with time between two extreme positions. The separation between these two extreme positions, in micrometer (μm), is _____.

PWO208

10. The maximum speed in $\mu\ m/s$ at which the 8th bright fringe will move is _____.

PWO209

For More Join: @IITJEE_Advanced

ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B	C	B	B	D	D	A	B	A	B
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	B	B	C	A	C	C	D	A	C	C
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	C	C	D	B	D	C	A	C	D	A
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	A	B	D	D	D	D	B	D	D	C
Que.	41	42	43	44	45	46	47	48	49	50
Ans.	D	B	C	B	B	C	D	A	D	C
Que.	51	52	53	54	55	56	57			
Ans.	D	B	A	C	A	A	C			

EXERCISE (O-2)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	ACD	BD	BCD	ABD	BC	CD	A	A	A	A
Que.	11	12	13							
Ans.	C	A	(a-ii), (b-iv), (c-i), (d-iii)							

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EXERCISE (S)

1. 52 nT-m

2. (a) 3.54×10^{-17} T, (b) 2.13×10^{-17} T

3. 48, 21, $\frac{32}{3}$, $\frac{9}{2}$, 0 $m.m.$;

4. 4

5. $\left(n - \frac{1}{48}\right)\lambda = x_1 - x_2$

6. 8 μm

7. 1100

8. 3

9. (a) 5.4 μm wide, (b) 0.99 mm high

10. 0.6mJ

11. 30°

EXERCISE - JEE (Main) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	4	2	2	3	1	3	1	1	4	462
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	2	3	2	0.20	1	10	3	2	2	3

EXERCISE - JEE (Advanced) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	3	BC	CD	C	A	BD	AD	8	601.50	24.00

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Important Notes

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